World models learn their simulator's integrator, not the physics it approximates

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Abstract

Probing asks whether a model represents a quantity by fitting a readout to it. Its known weakness is that accuracy mixes what the representation stores with what the probe can learn, and the accepted remedy is better probe hygiene. We show the remedy addresses the wrong failure. In a DreamerV3 world model trained only on pendulum video, a polynomial readout fitted to true physical energy reaches a correlation of 0.9999. The model's own transition does not preserve that direction, and enforcing it during imagination makes the model's physics worse. A label-free search tracks energy slightly less well, but the transition preserves it 6.7× better, and that search improves the physics by 42%. Whether a latent encodes a quantity and whether the dynamics preserve it are different properties. They dissociate on five further axes: untrained models, out-of-distribution states, a second architecture, an untrained transition, and actuation, with decodability surviving in every case where dynamical structure fails. The cleanest axis holds the data fixed: a different architecture on identical trajectories is 767× worse.

A statistic on the transition operator separates the two properties in every case, and we causally deploy the quantity it selects. Enforcing it cuts drift in decoded physical energy by 55–76% on 512 unseen trajectories, and 0 of 60 magnitude-matched random directions beat it. Setting it steers the imagined world's true energy, with rank correlation 0.80–0.91 and 0 of 75 controls beating it, and the effect survives an interchange 50 steps into autonomous imagination. We measure all of this on the model's own imagined rollouts against matched nulls.

Why the selected quantity differs from textbook energy has a candidate answer, and we state its status precisely. The data comes from a symplectic integrator, which conserves a shadow $\tilde{H} = H + \mathcal{O}(\Delta t)$ rather than H. On ground truth the trajectories conserve the quantity at the integrator's predicted coefficient 591× better than H, with nothing fitted, and correcting that term flips the intervention from harmful to helpful on every model. We cannot establish that the model has internalised it: a cross-evaluation control shows our one-step statistic depends on the trajectories we evaluate it on, and the instruments that would separate model from data are degenerate on these assets. The dissociation and its causal deployment do not rest on that account.

1 Introduction

Probing is how the field asks whether a model represents a quantity: fit a readout, report the accuracy. We train a DreamerV3 world model on pendulum video alone, and a polynomial readout fitted to the pendulum's true energy reaches $|\rho|_E = 0.9999$. By the usual reading, the model represents energy about as well as a model can.

It does not use it. The model's own transition fails to preserve the direction that readout identifies, and enforcing it during imagination makes the model's physics worse. A label-free search tracks energy less well, at $|\rho|_E = 0.973$, but the transition preserves it 6.7× better, and enforcing that one improves the physics by 42%. Whether a latent encodes a quantity and whether the dynamics preserve it are different properties, and no amount of probe hygiene closes the gap, because nothing is wrong with the probe.

We measure the second property directly, with a statistic on the transition operator. The two dissociate on six axes, and the model causally uses the quantity the operator selects rather than merely encoding it. Enforcing that quantity cuts drift in decoded physical energy by 55–76% on unseen trajectories. Setting it steers the imagined world's true energy, and the effect survives 50 steps into autonomous imagination.

Why should the two come apart at all, when we trained the model on pendulums? One candidate answer is that the training data does not conserve textbook energy either. It comes from a symplectic integrator, which conserves a nearby shadow $\tilde{H} = H + \mathcal{O}(\Delta t)$ (Hairer et al., 2006a). On ground truth that predicted coefficient is conserved 591× better than H with nothing fitted, and correcting the $\mathcal{O}(\Delta t)$ term flips the intervention from harmful to repairing on every model. We offer this as an account of the data, and Section 4 is explicit that our instruments cannot decide the corresponding claim about the model.

Why this matters for interpretability. A probe can be perfect and still wrong. We fit a polynomial readout to true physical energy, and it reaches $|\rho| = 0.9999$, as good as a probe gets. Yet the model's own transition preserves it 6.7× less well than the label-free scalar, and enforcing it during imagination makes the model's physics worse, while the label-free scalar improves it by 42%. Nothing is wrong with the probe. We fitted it to a quantity the generating process does not conserve, and no amount of probe hygiene (Hewitt and Liang, 2019; Belinkov, 2022) detects that, because probe hygiene tests the probe.

Measuring the operator instead. The discriminating measurement runs on the transition, not on the representation: we apply one autonomous step at a real encoded state and ask how much a candidate scalar moves (Equation 2). This single statistic separates trained from untrained models (74×) and conservative from dissipative (39×, no overlap). It also separates in-distribution from out (55–267×), one architecture from another (767×), and a label-free scalar from a supervised probe of the same quantity (6.7×). A probe reports success on every one of those contrasts, and the dissociation runs one way in each case we can construct: decodability survives where dynamical structure fails.

The model uses the quantity the operator selects. During imagination we project the latent back toward its level set, and on trajectories the method never saw that reduces drift in decoded physical energy by 55–76% at long horizon. No control beats it: 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions. Setting the scalar to an independent trajectory's value moves the imagined world's true energy toward that trajectory's (rank correlation 0.80–0.91, 0 of 75 controls). We apply the same interchange 50 steps into an autonomous rollout, at a state the model reached by itself, and it works nearly as well. A dormant pathway would not (Makelov et al., 2024).

Contributions. (i) A demonstration that probing over-reports, with a decisive control: a supervised probe of the true quantity, at $|\rho| = 0.9999$, identifies a direction the dynamics do not use. Enforcing it harms the model. The dissociation appears on six axes and runs one way in all of them, the cleanest holding the data fixed while changing the architecture (767×). (ii) An operator statistic that discriminates where probes cannot, with magnitude-matched rather than norm-matched controls. Prior work on this system used a norm-matched null, which controls nothing, because the correction is scale-invariant in the constraint. (iii) Causal evidence: the model deploys the selected direction, on 512 unseen trajectories and 50 steps into autonomous imagination, against magnitude-matched nulls that fail 0 of 60 and 0 of 75. (iv) An account of the gap, with its evidential status separated: the shadow Hamiltonian of the generating integrator explains why the perfect probe targets the wrong quantity, and we verify that on ground truth. We also report in full why the parallel claim about the model resists measurement here. (v) Stated boundaries: the correction confers no benefit to a planner and does not select checkpoints better than validation loss, for reasons we measure.

2 Recovering a latent invariant

Model and data. We analyze the published DreamerV3 architecture (Hafner et al., 2023): 13.5M parameters, categorical 32×32 stochastic latents, a convolutional encoder and decoder, KL balancing, and unimix. We train only the world model, with no actor or critic.

Data come from Gymnasium's Pendulum-v1 (Towers et al., 2024), which we render per step and block-average to 64×64. We set the simulator state directly, sampling $\theta \sim U[-1.8, 1.8]$ and $\dot{\theta} \sim U[-2.2, 2.2]$, and we reject any trajectory that reaches the $|\dot{\theta}| = 8$ clip, since clipping breaks conservation. Actions are zero unless stated, and a single frame does not reveal velocity, so the model must infer $\dot{\theta}$ from the sequence. We train on 204 trajectories of 120 frames for about 6,500 gradient steps, reserving 52 as a post-training analysis set. Three seeds pass checks fixed before analysis: KL above 1 nat, one-step decoding $4\times$ better than the dataset mean, and finite rollouts. Appendix A gives the rest. Our claims concern a DreamerV3 world model trained on pendulum video, not DreamerV3 as a full reinforcement-learning agent.

Latent coordinates. After training we freeze the model. Let h be the deterministic recurrent state on the analysis set and $\bar{h}$ its mean. We take the top 12 principal directions of h, drop any without support in the data, and let P be the resulting orthonormal map. All extraction and perturbation operate on $z = P^\top(h - \bar{h})$, and $F(z)$ is the projection through P of the model's one-step displacement $T(h) - h$. Using the model's own displacement avoids fitting a surrogate vector field and then analysing that instead.

The candidate family. We search degree-4 polynomials in z, writing $\varphi(z)$ for the monomials up to degree 4 in the 12 latent coordinates. A candidate is $C(z) = a^\top \varphi(z)$ over 1819 monomials, excluding the trivially conserved constant. Energy is quadratic in $\dot{\theta}$ and needs polynomial terms to approximate $\cos\theta$, and the latent is an unknown nonlinear encoding of $(\theta, \dot{\theta})$ rather than those coordinates. The family must therefore express energy through that distortion while staying small enough to fit.

The invariance criterion. An invariant is constant along a trajectory and differs between trajectories. We score a candidate by

$$\text{ratio}(C) \;=\; \frac{\text{mean within-trajectory variance of } C}{\text{total variance of } C}, \tag{1}$$

which is 0 for a perfectly conserved scalar and near 1 for one wandering as much within a trajectory as across the dataset. The denominator keeps it non-trivial: without it $C = 0$ would score perfectly while distinguishing nothing. Both variances are quadratic forms in a, so with W the mean within-trajectory covariance and T the total covariance this is a Rayleigh quotient $a^\top Wa / a^\top Ta$. Its minimiser is the smallest generalized eigenvector of $Wa = \lambda Ta$, and one eigendecomposition returns the whole family, ranked.

Selecting among conserved candidates. Conservation alone does not pick out a unique scalar. If C is conserved then so is any function of it, so E, E^2 and mixtures all sit near the top, and the leading eigenvector is generally a combination. We therefore fit C jointly with an antisymmetric operator B such that $F \approx B\nabla C$ with $B^\top = -B$, mirroring the Hamiltonian relation between a conserved quantity and the flow it generates (Arnold, 1989). Since $\nabla C^\top B \nabla C = 0$ for antisymmetric B, such a scalar is constant along the flow by construction. The criterion also separates candidates the ratio cannot: E^2 is exactly as conserved as E but does not pair with the same B. The fit is bilinear, so we alternate least-squares solves within the top eight eigenvectors, and the recovered C then has eight effective degrees of freedom rather than 1819. That matters on 52 trajectories, where a free fit could drive the in-sample ratio to zero by overfitting. Appendix A isolates the contribution of the flow criterion: conservation drives most of the recovery and intervention effect, while flow alignment mainly reduces variation across seeds.

Reference energy. We compare C against textbook pendulum energy $E = \frac{1}{6}\dot{\theta}^2 + 5\cos\theta$. Gymnasium integrates semi-implicitly, so it does not conserve E exactly. Section 4 shows

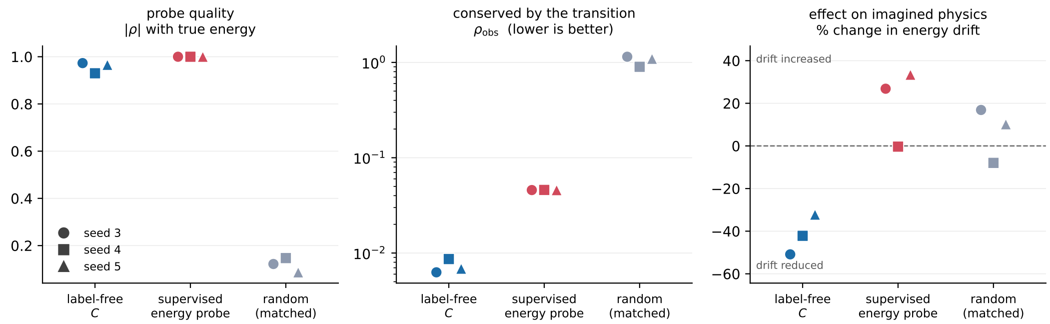

Figure 1: A perfect probe is not a conserved quantity. Colour is the constraint; marker is the model seed (all three shown, never averaged). Left: the supervised probe reads true energy almost perfectly. Centre: the model's own transition preserves it an order of magnitude worse than the label-free scalar. Right: acting on it fails to reduce rollout energy drift, while the label-free scalar cuts drift on every seed. Probe quality and dynamical relevance rank the three constraints differently.

what it conserves instead, and treats that mismatch as the crux rather than a caveat. The search sees only latent states and the model's own one-step transition, never θ, $\dot{\theta}$ or E. Ground truth enters afterward, when we score C against E. We selected the extraction dimension on three development models and fixed it before evaluating the three reported here.

3 Decodability is not dynamical structure

Establishing what the model conserves requires separating two things a probe conflates. A probe shows that a readout recovers a quantity from a representation; it says nothing about whether the model's own transition uses or preserves that quantity. We build the statistic that separates them, and we show that the two properties come apart. That separation is what lets Section 4 ask which quantity the transition actually preserves. We measure the second property directly, with a statistic on the transition operator. At a real encoded state we take one autonomous step and ask how much a candidate scalar changes,

$$\rho_{\text{obs}}(C) \;=\; \frac{\text{median}_t \big| C(T(z_t)) - C(z_t) \big|}{\text{std}_{\text{traj}}\, C}, \tag{2}$$

normalised so it is invariant to the scale of C and comparable across candidates. A value near zero means the transition preserves C; a value near one means a single step moves C by its entire across-trajectory spread.

A perfect probe identifies a direction the dynamics do not use. The sharpest test replaces the label-free search with the answer it is trying to find. We fit a degree-4 polynomial readout to true pendulum energy by ridge regression on the same latent, then use it to drive the identical correction. It reaches $|\rho|_E = 0.9999$: as a probe it is essentially perfect.

Figure 1 shows the outcome: the supervised probe tracks energy better than the label-free scalar on every seed, yet the transition preserves it 6.7× worse. The median ρ_{obs} is 4.58×10^{-2} against 6.9×10^{-3}, and 5.3–7.3× per seed; both hold on 3 of 3 models. It also never repairs the rollout. Its effect on energy drift is +26.8% and +33.4% on two models and −0.3% on the third, while the label-free scalar cuts drift on all three by a median −42.2% (32–51%). A magnitude-matched random constraint sits at $\rho_{\text{obs}} \approx 1.07$ and changes drift by +4.0%. The direction that best predicts the physical variable and the direction the model's dynamics actually respect are not the same direction.

This also answers the practical objection to a label-free method: the search is not a workaround for missing labels. It finds something a supervised fit on perfect labels does not, because it optimises conservation by the transition rather than agreement with the variable. Section 4 identifies what it finds, and why fitting true energy cannot find it.

Table 1: Decodability survives while dynamical structure degrades, across five independent axes. ρ_{obs} for the trained conservative model in-distribution is 6.3–8.7 $\times 10^{-3}$.

axis	contrast	decodability	ρ_{obs} degradation
training	untrained network	$\lvert\rho\rvert_E$ up to 0.908	74×
state space	outside training energies	free probe 0.999	55–267×
architecture	deterministic conv-GRU	$\lvert\rho\rvert_E$ 0.91 (2 of 3 seeds)	767×
objective	transition never trained	$\lvert\rho\rvert_E$ 0.97	∼10 000×
actuation	torque applied	$\lvert\rho\rvert_E$ 0.72–0.88	balance law absent

The dissociation is directional, and appears on five further axes. In every case we can construct, decodability survives where conservation fails; we have not found the converse. Table 1 collects them, and on each row a probe would report success.

The out-of-distribution row is cleanest, since it holds the model fixed and varies only the region of state space. Outside the training energy band a freely fitted probe still recovers energy at 0.999, so the encoder generalises. The transition conserves it 55–267× worse there, reaching values indistinguishable from a random constraint. The two properties come apart inside a single trained model, split by where in state space we ask.

The actuation row is the only axis testing a balance law rather than a conservation law; Appendix B gives it in full.

We have not established whether conservation alone predicts the intervention. The candidate set holds energy correlation fixed and spans five orders of magnitude of conservation. Within it, the rank correlation with repair is +0.19, +0.57 and +0.19 across seeds, and the 95% interval excludes zero on one of three. We report this as a negative, and the control is hard for a reason. On a system whose transition conserves energy the well-conserved latent directions are the energy-like ones, so a matched band exists only at low decodability, and on one seed we cannot build one at all. Appendix B gives the full analysis. What holds on all three seeds is that the recovered scalar repairs: −16.7%, −7.3% and −6.8%.

Where the phenomenon stops. Both boundaries state the same thing twice: the invariant extends no further than the training distribution and no further than RSSM-like architectures. On a deterministic conv-GRU, identification transfers on 2 of 3 seeds under both training budgets while conservation transfers on none, a 767× gap, and seven times more open-loop training moves the median barely. That comparison does not isolate architecture, since an open-loop reconstruction term and a prior–posterior KL are different losses; Appendix B gives both boundaries in full.

4 A candidate account: the integrator's shadow Hamiltonian

The transition preserves something, and the previous section shows it is not the optimal readout of true energy. We give a candidate account of what, and we are explicit about which half of it we can establish. On the data side the account is provable and we verify it. The model side asks whether the network has internalised the integrator's quantity, or whether our statistic instead reflects the trajectories it runs on. The instrument we have cannot decide that, and we say so. Nothing in Sections 3 or 5 depends on this section.

The data does not conserve textbook energy. Trajectories come from semi-implicit (symplectic) Euler,

$$\dot\theta_{n+1} = \dot\theta_n + \tfrac{3g}{2l}\sin(\theta_n)\,\Delta t, \qquad \theta_{n+1} = \theta_n + \dot\theta_{n+1}\,\Delta t. \tag{3}$$

A symplectic integrator does not conserve the continuous system's Hamiltonian H. It conserves a nearby shadow $\tilde H = H + \mathcal{O}(\Delta t)$ (Hairer et al., 2006a). Here the first-order correction is $\frac{\Delta t}{2}\{T, V\} \propto \dot\theta \sin\theta$, giving

$$\tilde H \;=\; E \;+\; c^\star\,\dot\theta\sin\theta, \qquad c^\star \;=\; \tfrac{\Delta t}{2}\,mg\,\tfrac{l}{2} \;=\; 0.125, \tag{4}$$

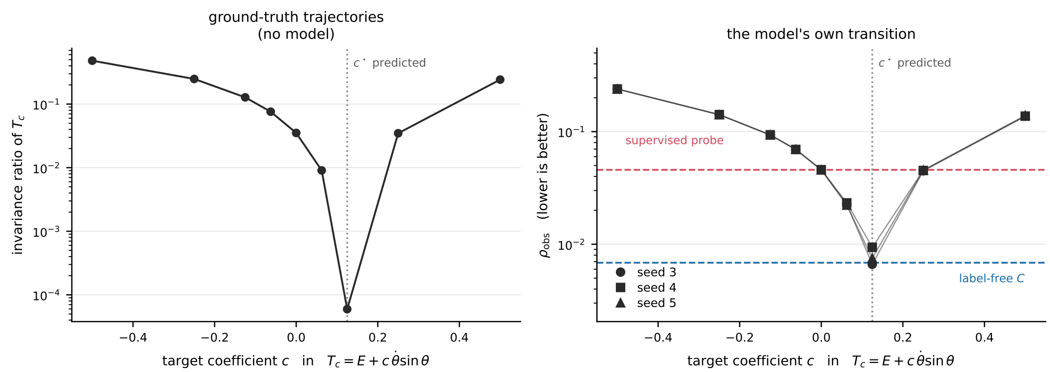

Figure 2: The model's transition conserves the integrator's shadow Hamiltonian, not textbook energy. Both panels sweep the target $T_c = E + c\,\dot{\theta}\sin\theta$; the dotted line is $c^\star = 0.125$ predicted from the integrator with no fitting. Left: on ground-truth trajectories, with no model involved, $T_{c^\star}$ is 591× better conserved than textbook energy. Right: the learned transition agrees, on all three seeds. The sweep passes from the supervised probe's level (red) down to the label-free scalar's (blue) exactly at $c^\star$. The wrong-sign control at $c = -c^\star$ is 12× worse (median over seeds), so this is not monotone in $|c|$.

with no free parameters. If the model learned the integrator's map rather than the textbook physics, its transition should preserve $\tilde{H}$, and a probe fitted to E then has the wrong target however well it fits.

A sweep, not a two-point test. A two-point comparison, a probe fitted to E against one fitted to $\tilde{H}$, would not settle this: $\tilde{H}$ is a different function, and its advantage could come from being easier to represent. We therefore sweep the family

$$T_c(\theta, \dot{\theta}) \;=\; E(\theta, \dot{\theta}) \;+\; c\,\dot{\theta}\sin\theta, \tag{5}$$

over a grid that contains $c^\star$, both signs, and wrong magnitudes 2× and 4× either way, refitting at each c with everything else held fixed. At $c = 0$ the sweep reduces to the supervised probe of Section 3 and reproduces its numbers to five decimals. The prediction is the location of a minimum, which no representational-ease artifact produces, and we registered it before the sweep ran.

The minimum is where the integrator says it is. Figure 2 shows both halves. On ground-truth trajectories, with no model involved, the invariance ratio of T_c falls by 591× between $c = 0$ and $c = c^\star$, which confirms the derivation and fixes the sign. On the learned transition, ρ_{obs} reaches its minimum at $c^\star$ on 3 of 3 models, falling 4.9–6.9× below its value at $c = 0$. The curve is not monotone in $|c|$: the wrong-sign, right-magnitude control at $c = -c^\star$ is 12× worse than at $c = +c^\star$ (range 10–14× over the three models). A post-hoc refinement over $c \in [0.090, 0.170]$ locates the minimum at 0.125 ± 0.005 on every seed.

The repair follows the same curve. Enforcing $T_{c^\star}$ during imagination changes energy drift by −49.5%, −31.5% and −29.3% on the three models, a repair on all three, while enforcing T_0, the perfect probe, gives +26.8%, −0.3% and +33.4%. Correcting an $\mathcal{O}(\Delta t)$ term in the target flips the intervention from harmful to helpful.

The coefficient tracks the simulator's timestep on the evaluation data. If the model has learned the integrator rather than the physics, $c^\star$ is a property of the discretisation and not a constant of the pendulum, so changing the simulator's timestep should move it: $c^\star(\Delta t) = \frac{\Delta t}{2} mg \frac{l}{2} = 2.5\,\Delta t$. We test this directly, generating four datasets at $\Delta t \in \{0.02, 0.035, 0.05, 0.08\}$ that are identical in every other respect and training three models on each. We express the sweep relatively, $r = c/2.5\Delta t$, so the prediction is a single value of r at every timestep.

Table 2: The recovered correction tracks the simulator's timestep across a 4× range. Three models per timestep; ρ_{obs} is the median across seeds.

Δt	predicted $c^\star$	$\arg\min_r$ per seed	ρ_{obs} at $r{=}0$	at $r{=}1$	separation
0.02	0.050	$+0.75, +0.75, +1.00$	0.0171	0.0166	1.03×
0.035	0.088	$+1.00 \times 3$	0.0231	0.0103	2.24×
0.05	0.125	$+1.00 \times 3$	0.0449	0.0079	5.72×
0.08	0.200	$+1.00 \times 3$	0.1188	0.0088	13.5×

Table 2 shows the outcome. The minimum sits exactly at the predicted $r = 1$ on 10 of 12 models, and the models beat the wrong-sign control on 12 of 12. We regress the recovered coefficient on Δt through the origin, as the theory requires, and get a slope of 2.484 ± 0.058 against a parameter-free prediction of 2.500, agreement to 0.6% with nothing fitted.

Two scalings make the mechanism visible. As the timestep coarsens, ρ_{obs} for textbook energy grows as $\Delta t^{1.41}$, while ρ_{obs} for the shadow quantity stays at the model's own floor, 0.008–0.017 across the whole range. The separation between them therefore grows from 1.03× to 13.5×, purely because textbook energy becomes worse conserved while the shadow quantity does not. At every timestep, the quantity best conserved along these trajectories is the integrator's rather than the physicist's.

We report one registered failure. Our preregistration required a slope within 20% and an intercept whose interval contains zero; the two-parameter fit gives a slope of 2.611 ± 0.110 but an intercept of -0.0072 ± 0.0057, which excludes zero. The smallest timestep drives it entirely, and a post-hoc finer grid puts the minimum there at $r = 0.875$ on all three seeds, a real 12% low bias rather than grid coarseness. That is the regime where the whole separation is only 1.03× and the residual curve is flat to within 2%.

What this does not settle. Table 2 is a statement about the data with the model in the loop, and we cannot cleanly remove the data. Our statistic ρ_{obs} measures how well a candidate is conserved along given trajectories, so a model trained at one timestep and a dataset generated at another both influence the argmin. A registered cross-evaluation control settles which dominates, and the answer is unfavourable: crossing models against datasets, the argmin follows the evaluation set in all six cells and the checkpoint in none (0/3 on both off-diagonals). The same control voids a companion experiment that appeared to identify the integration scheme; semi-implicit Euler and velocity Verlet reduce to the identical position recurrence $\theta_{t+1} = 2\theta_t - \theta_{t-1} + a(\theta_t)\Delta t^2$. They differ only in whether $\dot\theta$ carries a backward or a central difference label, and the pixels never show that variable.

Two alternative instruments fail as well. Measuring conservation along the model's own imagined rollout removes the evaluation trajectory entirely, but it has no power here. The sweep contrast is 1.12–1.45× against $\sim$5.7× for the one-step measure, and within a single model the argmin spans the full grid across horizons 2–100. A forced choice across timesteps is unreadable for a different reason. A model run at a timestep it never trained on is $\sim$90× worse in one-step error (0.007 → 0.6 rad), so its prediction error is nine times the separation between candidates. A pre-committed gate withheld what would otherwise have read as a clean positive.

So we have not shown the model-side claim false; we have shown it unsubstantiable with these assets. Separating model from data needs a model that is in distribution at more than one timestep, and we do not have one. What survives is the data-side fact, which is provable and which we verify. On ground truth the integrator's predicted coefficient is conserved 591× better than textbook energy with nothing fitted, and correcting that $\mathcal{O}(\Delta t)$ term flips the intervention from harmful to repairing on every model. That is enough to explain the perfect probe's failure: we had fitted the probe to a quantity the generating process does not conserve, and no probe hygiene detects this, because nothing is wrong with the probe. The target is wrong. It is not enough to claim the model learned the integrator, and we do not.

The residual is worth recording either way: median ρ_{obs} at $c^\star$ is 7.5×10^{-3} against the label-free scalar's 6.9×10^{-3}, a gap of $1.10\times$, so the shadow quantity accounts for essentially all of the $6.7\times$ gap in Section 3. The sweep is also internally decisive about the probing point, and that reading needs no model-versus-data attribution at all. At $c^\star$ the readout correlates with textbook energy worse than at $c = 0$ (0.977 against 0.9999), while it is conserved six times better and repairs the rollout instead of harming it. Correlation with the labelled quantity and fitness for the dynamics move in opposite directions along a single one-parameter family. The label-free method's advantage likewise does not depend on this section. It optimises conservation under the learned transition, so it does not need a target a human knows how to write down, and here it recovered an $\mathcal{O}(\Delta t)$ correction nobody supplied.

5 The privileged direction is causally deployed

If the recovered scalar were only a descriptive correlate, enforcing it would have no reason to help and setting it no reason to steer. Three interventions say otherwise, each one against matched controls that differ only in where they push, never in how hard. That distinction matters, since the obvious norm-matched null controls nothing (Appendix B).

Projecting the latent back toward its level set during imagination reduces drift by 55–76% at $H = 190$. We measure drift on decoded physical energy rather than pixels, over 512 trajectories from a disjoint generator seed, and we freeze the scalar and the coordinate frame. Among the controls, 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beat it.

Setting the scalar to an independent donor trajectory's value moves the imagined world's true decoded energy toward that trajectory's true energy (rank correlation 0.802–0.914, 0 of 75 controls). A preregistered offset sweep is monotone in both directions on every model. We then repeat the interchange 50 steps into an autonomous rollout, on a state the model reached by itself rather than one the encoder produced. It costs little (0.762–0.849 against 0.803–0.924 at depth 0, 0 of 13 controls), and a dormant pathway would not survive that (Makelov et al., 2024).

The same pipeline refuses when there is nothing to find. On matched dissipative models no nontrivial first integral exists, and the search returns a direction indistinguishable from random: 24 of 60 random draws beat it. Its operator statistic sits at 3.4–3.7×10^{-1}, the range of an untrained network and $39\times$ worse than any conservative model, with no overlap. Appendix B gives protocols and per-model numbers.

6 Limitations

This study covers two smooth Hamiltonian systems, a pendulum and a two-degree-of-freedom anharmonic oscillator, and neither has contacts or multiple interacting objects, nor have we tested discontinuous dynamics. Each arm has three trained models, and the model seed is the independent unit. Three is too few for a precise effect size, so we report ranges rather than confidence intervals. The extraction minimises an invariance ratio that is the normal component of one-step error, so a low normal error shows the optimiser meeting its objective, not a finding. The empirical content is that the resulting direction coincides with physical energy, which supervision does not reproduce. The architecture comparison likewise does not isolate architecture, because an open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths.

No downstream benefit, and we measured why. Both practical uses we tested fail, for one reason. We plan with CEM over the learned model on an energy-targeting task, and the model plans well, beating random actions by +1.65, 95% CI $[+0.50, +2.79]$. Yet no correction arm differs from no correction over three models and 20 episodes each, and the largest effect is 0.0014 against a return standard deviation of 0.575, or 0.24% of one episode's spread. All four registered predictions fail. The cause is structural: the correction acts on accumulated drift, and at the planner's horizon it shifts imagined energy by 0.3% of the spread across candidate action sequences. The planner ranks plans by that spread, so a shift this small cannot change which plan wins. We measure our repair over 100 autonomous steps, and a re-planning controller never imagines long enough to accumulate what the method removes. Looking further ahead does not help, because planning quality itself degrades with horizon. The same effect-size limit defeats drift as an online trust signal, which plain latent displacement beats on every model, and Appendix B gives both in full.

The method finds only what the data gave the model a reason to represent. The pendulum's rod length never changes, an exact constraint, yet a search for a residual $G(z) = 0$ fails, and the recovered direction correlates with the true residual at 0.004–0.032. Minimising $\mathbb{E}[G^2]$ is the wrong objective, ranking the true constraint 1525th of 1819 directions. The model barely encodes the constraint at all, and a probe predicts it from the latent at held-out $|\rho| = 0.19$. A quantity that never varies carries no information, so nothing in a reconstruction objective pushes it into the latent, and it can live in the decoder's weights. This bounds the approach to quantities that vary, and it is why we report no released-checkpoint experiment: limb lengths do not vary either.

Finally, the two-invariant case limits the extraction rather than the model. Where the system conserves both energy and angular momentum, the conserved subspace reliably contains both, at angular-momentum correlation 0.94–0.98, while the joint fit isolates either in only 1 of 7 measurements. Flow alignment selects a single direction inside that subspace, and it has no unique answer when two exist.

We cannot attribute the shadow correction to the model rather than the data. Section 4 establishes the account on the data side and reports, in the same place, why the model side resists measurement here. We compute ρ_{obs} along evaluation trajectories, and a registered cross-evaluation control shows the argmin follows the evaluation set in 6/6 cells and the checkpoint in 0/6. Of the two alternatives that remain, the rollout statistic has no power, at a contrast of 1.12–1.45×. The cross-timestep forced choice is unreadable, because a model evaluated off its training timestep is $\sim 90\times$ worse in one-step error. Deciding the question needs a model in distribution at more than one timestep. Until then the claim is unsubstantiated rather than refuted, and we construct the paper's other results so that they do not depend on it.

7 Related work

What probes establish. Probing shows a variable is recoverable, not that the computation uses it (Alain and Bengio, 2017). Control tasks (Hewitt and Liang, 2019) and later critiques (Belinkov, 2022) address the confound between what the representation stores and what the probe can learn. Ours is a different failure. We fit the probe in Section 3 to the true quantity and it reaches 0.9999, so no control-task correction applies, and it still identifies a direction the dynamics do not use. Work on emergent world representations goes beyond probing by intervening (Li et al., 2023; Nanda et al., 2023), the logic ours follows. Makelov et al. (2024) show such interventions can succeed through pathways the model does not normally use, and our depth-50 interchange answers that concern.

Discovering conserved quantities. ConservNet minimises a noise-regularised within-group variance (Ha and Jeong, 2021), and our invariance ratio is the closed-form Rayleigh-quotient analogue of that objective. AI Poincaré counts conserved quantities from trajectory manifolds (Liu and Tegmark, 2021), and ConCerNet learns a conservation law then projects the learned dynamics onto it (Zhang et al., 2023). FINDE finds first integrals and preserves them by projection (Matsubara and Yaguchi, 2023), and Noether Networks meta-learn a conserved quantity and apply it at test time, including on pixel pendulums (Alet et al., 2021). All impose or co-train conservation. We ask whether a model trained with no such term already has a scalar its transition preserves, and we measure that on the frozen transition rather than building it in.

Structure by design, and whether it is what helps. Hamiltonian and Lagrangian networks build conservation into an architecture (Greydanus et al., 2019; Cranmer et al., 2020). Gruver et al. (2022) find HNN generalisation comes from modelling acceleration and coordinate choice rather than from symplectic structure, which licenses the objection that our projection is a generic regulariser. The magnitude-matched null answers it: with step size fixed, 0 of 60 random and 0 of 15 equal-norm tangent directions match the recovered one. On dissipative models the pipeline returns a direction indistinguishable from random.

Drift, and evaluating physics in world models. Accurate one-step models drift over long rollouts, and several fields document it; in machine-learned force fields, force accuracy and simulation stability decouple sharply (Fu et al., 2023). Projection onto a level set is a standard structure-preserving device (Hairer et al., 2006b), and our correction applies it in a learned latent, to a discovered rather than a given invariant. Evaluations of world models otherwise measure prediction quality or control performance (Hafner et al., 2023; Ha and Schmidhuber, 2018; Schrittwieser et al., 2020), and physics benchmarks evaluate generated output; neither of those reads the internal state. The disentanglement literature (Locatello et al., 2019) documents how readily an appealing structural claim about representations survives without the control that would falsify it.

8 Conclusion

A probe can read a quantity out of a world model almost perfectly while the model's own dynamics ignore it. In a DreamerV3 trained only on pendulum video, a readout fitted to true energy reaches $|\rho| = 0.9999$, and enforcing it harms the imagined physics, while a label-free direction that the transition actually preserves repairs it. Decodability and dynamical structure are separate properties of a representation, and probing measures only the first.

We can say why the target was wrong without being able to say what the model internalised. The training data comes from a symplectic integrator and does not conserve textbook energy either. On ground truth the trajectories conserve the integrator's predicted quantity 591$\times$ better than H with nothing fitted, and applying that correction flips the intervention from harmful to repairing on every model. The stronger reading, that the network has learned the numerical scheme, we could not establish. Our one-step statistic depends on which trajectories we evaluate it on, and the two instruments that would remove that dependence are respectively powerless and unreadable on these models. We report that in full, because the alternative is a headline the measurement does not carry.

None of the causal results rest on it. A probe fitted to true energy at $|\rho| = 0.9999$ identifies a direction the dynamics do not use, and enforcing it degrades the model's imagined physics. The cause is not a poor probe: we fitted it to a quantity the generating process does not conserve, and no probe hygiene detects that. Decodability and dynamical structure are different properties that come apart in every setting we could construct, and only a measurement on the transition operator tells them apart.

The practical consequence is narrow. That a physical variable is decodable says little about whether the model's rollouts respect it. It says least where it matters most: out of distribution, in architectures that predict well without conserving, and when the variable comes from supervision rather than from the dynamics. The operator statistic costs one autonomous step per state, and on this evidence it does not buy a better controller.

A Experimental details and extraction ablations

A magnitude-matched null, and why the obvious one fails. The natural correction projects the latent back toward the level set, $z \leftarrow z - \alpha(C(z) - C_0)\,\nabla C/\|\nabla C\|^2$. This is a Newton step, so its size grows with the size of the violation, and random constraints show far larger violations than conserved ones. Matching random draws in coefficient norm controls nothing, because the step is invariant under $C \to \lambda C$. Measured, random draws take steps 29$\times$ larger than the recovered constraint, so the comparison conflates direction with magnitude.

We therefore fix the step size and vary only direction, $z \leftarrow z - \varepsilon\,\mathrm{sign}(C - C_0)\,\nabla C/\|\nabla C\|$, over a frozen ε grid. Arms then differ in where they push, never in how hard.

Three kinds of latent trajectories. We use three latent trajectories and keep them separate throughout.

An observation-conditioned trajectory is the sequence of deterministic recurrent states produced when the model receives the next real frame at every step. This is what the encoder returns on the analysis set.

The one-step transition T is the model's autonomous recurrent update with zero action and no new observation.

An imagination rollout starts from one encoded state and repeatedly applies T without supplying any later frames.

We estimate the latent flow by applying T once at each state of an observation-conditioned trajectory, so the transition is autonomous even though the states come from real video. Section 3 measures conservation along observation-conditioned trajectories. Section 5 tests what happens to the same scalar under autonomous imagination.

Code and data. The extraction code, the run logs behind every figure and table, and the source of this paper are at https://github.com/Zarand3r/world-model-invariants. The figures regenerate from the committed logs without a GPU.

Preliminary experiments. We developed the extraction procedure in preliminary experiments on smaller recurrent models. Those experiments motivated the untrained, dissipative, and intervention controls used here, and we do not use them as evidence for the DreamerV3 results.

Training. We set free bits to 0 rather than DreamerV3's reference default of 1.0. The measured KL divergence settles near 1.8 nats, well above collapse, so the floor is inactive here. All three trained models pass the acceptance checks; none were discarded.

Data split. Training uses trajectories 0–203 of the 256-trajectory dataset. The remaining 52 form the analysis set. We fit C on those 52 trajectories and evaluate the Section 5 correction on the same 52, so the absolute effect is in-sample with respect to invariant fitting. The recovered and random-constraint arms use the same trajectories.

Scoring the correction over a grid. We score the slope of rollout error over the fixed α grid rather than the minimum on that grid. Taking the minimum selects the most favourable of five points after seeing the curve. In one diagnostic run that rule would have reported a 5.9% improvement for an arm whose error rose monotonically to twice baseline.

Flow alignment contributes reproducibility. Holding the candidate family, data, dimension, degree, and α grid fixed, we select C either by the invariance criterion alone or by the joint criterion. Conservation alone recovers energy at 0.950 against 0.973 for the joint rule, and gives a slightly better mean intervention slope (-0.087 against -0.077, winning on two of three models). The difference is in spread: recovery varies by 0.035 across seeds under conservation alone and by 0.008 under the joint rule.

The pairing residual does not track recovery. As the extraction dimension rises from 6 to 12, energy recovery improves from 0.721 to 0.967 while the pairing residual worsens from 0.738 to 0.876. Recovery already reaches 0.967 at dimension 8, where the residual is lower at 0.813. The added dimensions do not carry energy while resisting the Hamiltonian fit: directions 7–12 account for 44.9% of the residual and 43.5% of the flow. Figure 4 gives the sweep.

B Causal-intervention protocols and per-model numbers

If the recovered scalar is only a descriptive correlate, enforcing it has no reason to help and setting it has no reason to steer. Three interventions test the alternative, each with matched controls.

A magnitude-matched null. We fix the step size and vary only direction, $z \leftarrow z - \varepsilon\,\mathrm{sign}(C - C_0)\,\nabla C/\|\nabla C\|$, over a frozen ε grid, so arms differ only in where they push, never in how hard. The obvious alternative is a level-set projection with random draws matched in coefficient norm, and it controls nothing, because the step is invariant under $C \to \lambda C$. Measured, random draws then take steps 29× larger than the recovered constraint. Appendix A gives the derivation.

Correcting the invariant repairs the imagined physics. We evaluate on decoded physical energy rather than pixels. A non-learned geometric readout recovers θ by image moments and $\dot{\theta}$ by backward differences, matching the simulator's semi-implicit convention, with median decoded-energy error 1.6% of the across-trajectory spread. The statistic is secular drift of decoded energy over a rollout.

At $H = 100$ the recovered direction reduces drift by 32–51% across three models. On 512 trajectories from a disjoint generator seed, with C and the entire coordinate frame frozen, the effect does not shrink: −54.6%, −58.8%, −75.9% at $H = 190$. Across both horizons 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beat the recovered one on any model.

Setting the invariant steers the imagined physics. Repair shows that drift is costly. It does not show that C is a control variable. We move a recipient state minimally along the local C-normal until $C(z) = C(z_{\mathrm{donor}})$ for an independent donor trajectory, then let the rollout run free. We read the target from the model's own latent, never from true energy. The imagined world's true decoded energy moves toward the donor's true energy, rank correlation 0.802–0.914 across three models. Of the controls, 0 of 75 random and tangent directions beat it, and a preregistered offset sweep is monotone in both directions on every model.

The model's own dynamics use the subspace, not only states the encoder supplies. A subspace edit can produce the expected output change through a pathway the model does not normally use (Makelov et al., 2024), so we repeat the donor interchange 50 steps into an autonomous rollout, on a state the model reached by itself. The effect barely degrades: rank correlation is 0.762–0.849 at depth 50 against 0.803–0.924 at depth 0, 0 of 13 controls beat it at either depth, and equal-norm tangent edits stay near zero throughout. A dormant pathway would lose its effect precisely where this test applies it.

The same pipeline refuses when there is nothing to find. On matched dissipative models, where no nontrivial first integral exists, the extraction returns a direction we cannot distinguish from a random one. Random draws beat it 24 of 60 times, and it harms rollouts on 2 of 3 models. Its operator statistic sits at $3.4\text{–}3.7 \times 10^{-1}$, in the same range as an untrained network and 39× worse than any conservative model, with no overlap between the two groups.

Downstream tests in full.

No downstream benefit, and we measured why. Both practical uses we tested fail, for one reason. We plan with CEM over the learned model on an energy-targeting task and score return from the simulator's true state. The model plans well, beating a random-action policy by +1.65, 95% CI [+0.50, +2.79]. But no correction arm differs from no correction over three models and 20 episodes each, including an arm supplied with the action's power as a source term. The largest effect is 0.0014 against an across-episode return standard deviation of 0.575, 0.24% of one episode's spread, and all four registered predictions failed. The cause is structural. The correction acts on accumulated drift. At the planner's horizon it shifts imagined energy by 0.3% of the spread across candidate action sequences, and that spread is the quantity the planner ranks plans by, so the correction cannot change the planner's choice. We measure the repair over 100 autonomous steps, and a controller that re-plans every step never imagines long enough to accumulate what the method removes. Looking further ahead does not rescue it. Tripling the horizon leaves every arm indistinguishable (largest effect 0.007 against a return standard deviation of 0.862) while planning quality itself degrades, −0.34 to −0.56 in mean return. The correction would have something to correct only at horizons where the planner is already worse off.

The same effect-size limit defeats the other use. Accumulated drift at rollout step 25 predicts decoded energy error at step 100 at Spearman −0.02, +0.09 and −0.24. Plain latent displacement reaches +0.27, +0.29 and +0.31, a random-constraint control beats the drift predictor on two of three, and the registered prediction went the other way on every model. Causal deployment in imagination does not imply that the drift of a conserved quantity is a useful online signal. The effect on pixel error alone is also small (1.7% at H=50); the large effects here are on decoded physical energy, a metric we introduce because it measures the property in question.

The method finds only what the data gave the model a reason to represent. The pendulum's rod length is fixed, so the rendered bob holds its distance from the pivot to 0.67%, an exact constraint. Searching for a residual $G(z) = 0$ rather than a conserved scalar fails: the recovered direction correlates with the true residual at 0.004–0.032. We measured two causes. Minimising $\mathbb{E}[G^2]$ is the wrong objective, and it ranks the true constraint 1525th of 1819 directions. The model also barely encodes the constraint: a probe predicts it from the latent at held-out $|\rho| = 0.19$, with a residual four times its own variation. A quantity that never varies carries no information, so nothing in a reconstruction objective pushes it into the latent. It can live in the decoder's weights. This bounds the approach to quantities that vary, and it is why we report no released-checkpoint experiment: limb lengths do not vary either.

The two-invariant case limits the extraction rather than the model. On a system conserving both energy and angular momentum the conserved subspace reliably contains both (angular-momentum correlation 0.94–0.98 in every measurement), while the joint fit isolates either in only 1 of 7 measurements. The flow-alignment criterion selects a single direction inside that subspace, and has no unique answer when two independent invariants exist.

Conservation at matched decodability, in full.

We have not established whether conservation alone predicts the intervention. The direct control asks whether, among candidates matched in energy correlation, better conservation alone produces a larger repair. The 2-DoF system admits such a band. On two of three seeds, 397 and 398 of 400 candidates lie within 0.10 of the recovered scalar's $|\rho|_E$, and they span five orders of magnitude of conservation quality at $|\rho|_E \leq 0.11$. Across 20 candidates stratified by conservation, the rank correlation with repair is +0.19, +0.57 and +0.19 on the three seeds, with the 95% confidence interval excluding zero on one of three. We report this as a negative result: the relationship is visible but not established at $n = 3$.

Two further facts belong with it. The point estimate moves with the size of the candidate pool, $+0.29 \rightarrow +0.19$ on one seed between pools of 150 and 400. The statistic is therefore not stable to a design choice that should be immaterial, though the one-of-three verdict is identical under both pools. On the third seed we cannot build the band at all. Its reference $|\rho|_E$ is 0.111, only 19 of 400 candidates fall within 0.10 of it, and raising the pool from 150 to 400 adds none.

That last failure is the informative part. On a system whose transition conserves energy, the well-conserved latent directions are the energy-like ones. This is why we could not construct E10's original high-decodability form on the pendulum at any pool size, why the matched band here exists only at low decodability, and why on one seed it does not exist at all. This design may not separate the two properties on any system where the model has learned the physics. That is a statement about the difficulty of this control, not about the dissociation itself, for which the supervised probe above and Section 4 give direct evidence. One thing holds across all three seeds: the recovered scalar repairs, at −16.7%, −7.3% and −6.8%.

Under actuation the same split appears against a different law. The four axes above vary how well a conserved quantity survives; a fifth varies the law itself. We train three further models on the same pendulum under piecewise-constant random torque, where energy instead obeys $\mathrm{d}E/\mathrm{d}t = \tau\dot{\theta}$. They demonstrably use their actions: a 20-step open-loop rollout under the true action sequence beats one under the same actions shuffled across trajectories, 0.740–0.761 on 3 of 3 seeds. The unchanged extraction recovers an energy-like scalar on every model ($|\rho|_E = 0.72, 0.88, 0.86$), yet its motion under the model's own transition does not follow the balance relation. With the torque known and $\dot{\theta}$ from the same decoded readout, the rank correlation with the predicted source term is 0.041, 0.076 and 0.065. The power term accounts for 0.31% of the variance in ΔC, and the observed change is 4.5–5.4× larger than predicted. The pairing beats a shuffled-power control 3 of 3, so it carries something, but almost nothing. The model represents the quantity and does not implement the relation it obeys. That is also why a correction built on that scalar confers no control benefit (Section 6): there is no balance structure for it to enforce.

Where the phenomenon stops. Both boundaries above are the same statement seen twice. The invariant is bounded to the training distribution. Outside the training energy band a freely fitted probe still recovers energy at 0.999, while the transition conserves it 55–267× worse and reaches values indistinguishable from a random constraint. It is also bounded to the architecture. On a deterministic convolutional GRU with no stochastic latent, no KL term and no unimix, identification transfers on 2 of 3 seeds under both training budgets while conservation transfers on none. The gap is 767×: ρ_{obs} of 4.83–5.55 against the RSSM's $\sim 7 \times 10^{-3}$. Seven times more open-loop training moved the median from 5.33 to 5.26. This comparison does not isolate architecture. An open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths. The supportable scope of every claim here is therefore RSSM-like models, in distribution.

C Latent sensitivity analyses

For each direction u of the extracted subspace we measure its variance and the damage a displacement along it does to an imagination rollout,

$$V(u) = \mathrm{Var}(u^\top z), \qquad D_H(u) = \mathbb{E}\big[L_H(z + \epsilon u) - L_H(z)\big], \tag{6}$$

with L_H the H-step imagination error against the analysis set.

Calibrating the displacement. At $\epsilon = 0.05|z|$ the damage is about -0.5% of baseline with the sign varying across directions. Imagination is deterministic, so repeated unperturbed rollouts are identical and these values are not sampling noise. Damage for the leading direction runs -0.5%, $+3.1\%$, $+18.1\%$, $+46.4\%$, and $+128.8\%$ at $\epsilon/|z|$ of 0.05, 0.10, 0.25, 0.50, and 1.00. We use $\epsilon = 0.25|z|$, the smallest displacement whose damage exceeds 10% of baseline while the response still grows smoothly.

A displacement does not decay under imagination. After 40 steps it retains 6.3× its initial size at $\epsilon = 0.05|z|$, 2.8× at $0.25|z|$, and 1.9× at $0.5|z|$, as medians over the three models.

Variance and sensitivity at different horizons. At a 40-step horizon the two rankings anticorrelate on all three models (-0.364, -0.259, -0.406), with damage spanning 13.5× to 118.8× between the most and least consequential direction (Figure 3a). The ranking replicates: split-half rank correlation of D_H has median 0.95 across the settings we tried, with a minimum of 0.52 at the smallest displacement and longest horizon. The disagreement with variance does not extend to long horizons. Pooling three models and three displacements, median $\rho(V, D_H)$ is -0.252 at horizon 20, negative in 9 of 9 settings; -0.196 at horizon 50, negative in 5 of 9; and -0.028 at horizon 100, negative in 5 of 9, with one setting reaching $+0.727$. Once a rollout has diverged, high-amplitude directions dominate the error and the rankings realign.

The correction pushes along higher-damage directions at $+0.385$, $+0.692$, and $+0.287$ (Figure 3b), without having been given that ranking. The correlations stay well below 1, so the correction is only partly concentrated there. All of this is measured on the conservative models.

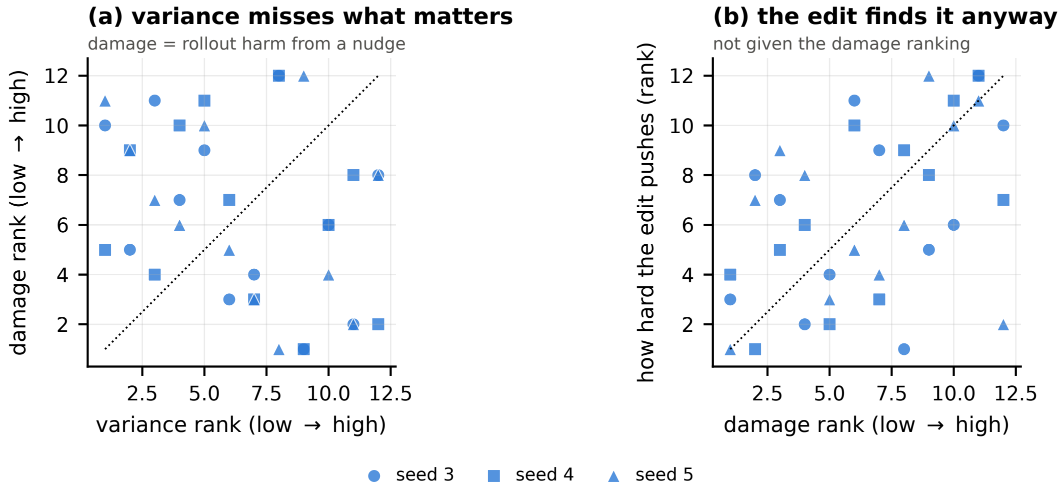

Figure 3: Each point is one of the 12 extracted directions, ranked within its own model; the dotted line marks equal ranks. (a) Variance rank against damage rank at a 40-step horizon. (b) Damage rank against the magnitude of the invariant correction along that direction.

Basis-independent properties reproduce better than coordinate selectors. The three models agree closely on quantities that need no choice of basis: energy recovery varies by 0.008, intervention slopes by 0.016, and the top-3 eigenvalue mass of $M_C = \mathbb{E}[gg^\top/\|g\|^2]$ with $g = \nabla C$ by 0.046 (0.448, 0.441, 0.487). Selectors on individual coordinates are less stable: ranking coordinates by ∇C and keeping the top three gives intervention slopes of -0.075, $+0.016$, and -0.089, beating a variance ranking on two models and changing sign across seeds. An orthogonal projector onto the top eigenspace of M_C beats the variance selector on all three models, though on one it sits at the 41st percentile of 100 random rank-3 subspaces. Since the correction is rank one at each state, the spectrum of M_C measures how much ∇C rotates as the state changes, and the amount is similar across models. This is consistent with the models encoding the same scalar in different latent coordinate systems, although we do not directly align their representations.

D Additional sweeps

Energy outside the highest-variance directions. On two of three models, principal directions 7–12 carry about twice the energy information of the top six (2.23× and 2.13×); on the third the ratio reverses to 0.91. We report the models separately because a median would hide the sign change. Figure 5 shows the decomposition.

Accumulated invariant error. Accumulated invariant error predicts which rollouts degrade most: an integral predictor reaches $R^2 = 0.187$ against 0.021 for a fitted power law in time at the horizon where both perform best, and shuffling destroys the relation. The current experiments do not distinguish weighted from unweighted accumulation. In a nested comparison with 400 bootstrap resamples, the weighted term adds explanatory power beyond the unweighted term in 1 of 9 seed–horizon combinations, and the unweighted beyond the weighted in 0 of 9, at $n = 512$.

References

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In ICLR Workshop, 2017.

Ferran Alet, Dylan Doblar, Allan Zhou, Joshua Tenenbaum, Kenji Kawaguchi, and Chelsea Finn. Noether networks: Meta-learning useful conserved quantities. In Advances in Neural Information Processing Systems, 2021.

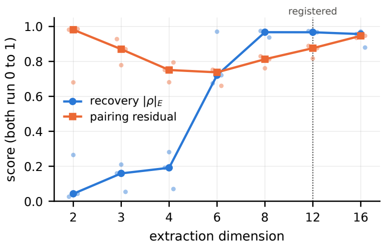

Figure 4: Energy recovery and pairing residual against extraction dimension. Lines are medians over the three models; points are individual models. Both quantities are dimensionless.

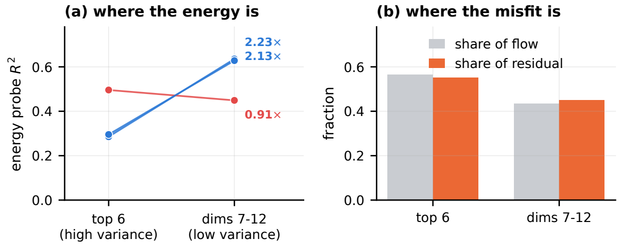

Figure 5: (a) Energy-probe R^2 for the top six principal directions against directions 7–12, per model. (b) Fraction of latent flow and pairing residual carried by each group, medians over three models.

V. I. Arnold. Mathematical Methods of Classical Mechanics. Springer, 1989.

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. Computational Linguistics, 48, 2022.

Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks. In ICLR Workshop, 2020.

Xiang Fu, Zhenghao Wu, Wujie Wang, Tian Xie, Sinan Keten, Rafael Gomez-Bombarelli, and Tommi Jaakkola. Forces are not enough: Benchmark and critical evaluation for machine learning force fields with molecular simulations. Transactions on Machine Learning Research, 2023.

Samuel Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In NeurIPS, 2019.

Nate Gruver, Marc Finzi, Samuel Stanton, and Andrew Gordon Wilson. Deconstructing the inductive biases of hamiltonian neural networks. In International Conference on Learning Representations, 2022.

David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In NeurIPS, 2018.

Seungwoong Ha and Hawoong Jeong. Discovering conservation laws from trajectories via machine learning. Physical Review Research, 3(4):L042035, 2021.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. arXiv:2301.04104, 2023.

Ernst Hairer, Christian Lubich, and Gerhard Wanner. Geometric Numerical Integration. Springer, 2006a.

Ernst Hairer, Christian Lubich, and Gerhard Wanner. Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations. Springer, 2nd edition, 2006b.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In EMNLP, 2019.

Kenneth Li, Aspen K Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. In ICLR, 2023.

Ziming Liu and Max Tegmark. Machine learning conservation laws from trajectories. Physical Review Letters, 126(18):180604, 2021.

Francesco Locatello, Stefan Bauer, Mario Lucic, Gunnar Rätsch, Sylvain Gelly, Bernhard Schölkopf, and Olivier Bachem. Challenging common assumptions in the unsupervised learning of disentangled representations. In ICML, 2019.

Aleksandar Makelov, Georg Lange, and Neel Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In International Conference on Learning Representations, 2024.

Takashi Matsubara and Takaharu Yaguchi. FINDE: Neural differential equations for finding and preserving invariant quantities. In International Conference on Learning Representations, 2023.

Neel Nanda, Andrew Lee, and Martin Wattenberg. Emergent linear representations in world models of self-supervised sequence models. In BlackboxNLP, 2023.

Julian Schrittwieser et al. Mastering atari, go, chess and shogi by planning with a learned model. Nature, 588, 2020.

Mark Towers et al. Gymnasium: A standard interface for reinforcement learning environments. arXiv:2407.17032, 2024.

Wang Zhang, Tsui-Wei Weng, Subhro Das, Alexandre Megretski, Luca Daniel, and Lam M. Nguyen. ConCerNet: A contrastive learning based framework for automated conservation law discovery and trustworthy dynamical system prediction. In International Conference on Machine Learning, 2023.